



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

END-SEMESTER EXAMINATION, SEMESTER II- 2015/2016

MT 105 – REAL ANALYSIS I

Time Allowed: Three hours

Answer all questions.

1. (a) Prove that $\sqrt{6}$ is an irrational number.

Deduce that $\sqrt{2} + \sqrt{3}$ is also an irrational number.

- (b) Find the supremum, infimum, maximum and minimum of each of the following subsets of $\mathbb{R}$, if they exist. If not, write “does not exist”.

(i) $(1, 2] \cup (3, 4]$

(iii) $\{x : 0 < x < 1\}$

(ii) $\{x \in \mathbb{Q} : x^2 < 3\}$

(iv) $\left\{ \frac{1}{1+x^2} : x \in \mathbb{R} \right\}$

2. (a) Using the $\varepsilon - N$ definition, prove that $\lim_{n \rightarrow \infty} \frac{n^2 + n - 1}{3n^2 - 4n - 1} = \frac{1}{3}$.

- (b) Let (x_n) and (y_n) be two convergent sequences of real numbers such that

$$\lim_{n \rightarrow \infty} x_n = l \text{ and } \lim_{n \rightarrow \infty} y_n = m. \text{ Using the } \varepsilon - N \text{ definition, prove that}$$

$$\lim_{n \rightarrow \infty} x_n y_n = l m.$$

3. Let (x_n) be a sequence of real numbers. Prove that if (x_n) is increasing and bounded above, then it is convergent.

Let (x_n) be the sequence defined by $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + x_n}$ for $n = 1, 2, \dots$.

Show that

- (i) (x_n) is an increasing sequence,

- (ii) $x_n \leq 2$ for $n = 1, 2, \dots$.

Deduce that (x_n) is convergent and find $\lim_{n \rightarrow \infty} x_n$.

Find $\sup \{x_n : n \in \mathbb{N}\}$ and $\inf \{x_n : n \in \mathbb{N}\}$.

Also, find $\lim_{n \rightarrow \infty} (x_n^2 - x_n - 2)$.

[Turn over]

4. (a) Using the $\varepsilon - \delta$ definition, prove that $\lim_{x \rightarrow 2} \frac{x^2}{3(x+2)} = \frac{1}{3}$.

(b) Prove that $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

5. (a) Let $f(x) = \begin{cases} x & \text{if } x \text{ is rational,} \\ -x & \text{if } x \text{ is irrational.} \end{cases}$

Show that f is continuous at $a \in \mathbb{R}$ if and only if $a = 0$.

(b) State and prove Rolle's Theorem.

State and deduce the Mean-Value Theorem. Verify the Mean-Value Theorem for the function given by $f(x) = x^3 + 2x^2 - x + 3$ on the interval $[0, 2]$.

6. Find each of the following limits:

(i) $\lim_{x \rightarrow 0} \left(\frac{3x - 2 \sin x}{x^3 \cos^2 x} \right),$

(ii) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right).$
